Blessed G/Michael Catholic Secondary School

Semester 1 Agriculture Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Leguminous crops enrich soil fertility primarily by fixing atmospheric nitrogen through symbiotic Rhizobium bacteria in their root nodules.

____ **2.** Drip irrigation has the lowest water-use efficiency among all contemporary irrigation methods.

____ **3.** Soil salinization occurs frequently in arid and semi-arid regions characterized by poor drainage and excessive surface evaporation.

____ **4.** Colostrum is the first milk produced by mammals after giving birth, containing concentrated antibodies essential for neonatal passive immunity.

____ **5.** Integrated Pest Management (IPM) advocates the total eradication of all pest insects exclusively using broad-spectrum synthetic chemical pesticides.

II. Choose the best answer among the given alternatives.

____ **6.** Which major cereal crop is indigenous to Ethiopia and serves as the primary staple grain for making injera?

A. Wheat (Triticum aestivum) **B.** Teff (Eragrostis tef)

C. Maize (Zea mays) **D.** Barley (Hordeum vulgare)

____ **7.** What is the primary visual symptom of severe nitrogen (N) deficiency in cereal crops?

A. Purplish discoloration on the lower leaf margins
B. Interveinal chlorosis on the youngest top leaves
C. General yellowing (chlorosis) starting on older lower leaves along the midrib
D. Marginal scorching and necrotic brown leaf tips

____ **8.** Phosphorus deficiency in plants is typically characterized by:

A. Purplish or reddish discoloration on stems and older leaves
B. Bleaching of flower petals
C. Stunted root growth with white spots
D. Rapid wilting under full sunlight

____ **9.** Which soil order is prevalent in the central highlands of Ethiopia, known as heavy black clay soils that crack when dry and expand when wet?

A. Vertisols **B.** Nitisols **C.** Arenosols **D.** Ferralsols

____ **10.** What broadbed maker (BBM) technology was specifically developed by Ethiopian agricultural researchers to manage Vertisol soils?

A. To drill deep boreholes for groundwater extraction
B. To drain excess surface water and construct raised seedbeds for early planting
C. To apply liquid pesticides automatically
D. To crush rocks in steep mountainous terrains

____ **11.** The practice of growing two or more crops simultaneously on the same piece of land in distinct rows is known as:

A. Monoculture **B.** Intercropping

C. Shifting cultivation **D.** Fallowing

____ **12.** Which of the following describes the ruminant stomach compartment where microbial fermentation of cellulose occurs?

A. Abomasum (true stomach) **B.** Rumen

C. Omasum **D.** Reticulum alone

____ **13.** Which stomach compartment in ruminants secretes hydrochloric acid and gastric enzymes, functioning similarly to the monogastric stomach?

A. Rumen **B.** Reticulum **C.** Omasum **D.** Abomasum

____ **14.** Silage is preserved green forage feed prepared through which biological process under anaerobic conditions?

A. Lactic acid fermentation
B. Aerobic mold composting
C. Thermal sun drying
D. Alkaline neutralization

____ **15.** Which fungal disease represents one of the most destructive threats to wheat production in the Ethiopian highlands?

A. Stem rust (Puccinia graminis f. sp. tritici)
B. Downy mildew of onion
C. Coffee berry disease
D. Bacterial wilt of enset

____ **16.** Enset (Ensete ventricosum), commonly termed the 'false banana', is a drought-resilient multi-purpose staple crop primarily cultivated in:

A. The arid Somali lowlands
B. The Southern and Southwestern regions of Ethiopia
C. The Afar depression
D. The Gambella wetlands

____ **17.** What devastating plant disease threatens Enset cultivation and is caused by the bacterium Xanthomonas vasicola pv. musacearum?

A. Bacterial Wilt of Enset (Xanthomonas wilt)
B. Coffee Leaf Rust
C. Late Blight
D. Smut disease

____ **18.** Soil erosion caused by rainwater runoff can be effectively mitigated on steep hillslopes through:

A. Deep contour plowing, terracing, and vegetative grass strips
B. Intensive downhill tillage along the slope
C. Overgrazing hillsides to compact topsoil
D. Complete deforestation of hillside trees

____ **19.** Which of the following organic amendments is most effective for improving both water retention in sandy soils and aeration in clay soils?

A. Urea chemical fertilizer
B. Decomposed compost or farmyard manure (organic matter)
C. Ground limestone (calcium carbonate)
D. Sodium chloride crystals

____ **20.** The economic principle stating that adding successive units of a variable input (such as fertilizer) to fixed land eventually yields smaller additions to output is:

A. The Law of Comparative Advantage
B. The Law of Diminishing Marginal Returns
C. The Law of Supply and Demand
D. The Economies of Scale

____ **21.** Which artificial insemination (AI) procedure is standard in commercial dairy cattle breeding?

A. Vaginal natural mating only
B. Recto-vaginal cervical deposition using cryopreserved straw semen
C. Surgical embryo implantation in the abomasum
D. Spontaneous hormone induction without semen

____ **22.** What term denotes the removal of unthrifty, diseased, or unproductive livestock from a farm herd to enhance herd genetic quality and profitability?

A. Castration **B.** Culling **C.** Docking **D.** Dehorning

____ **23.** Which of the following feeds is categorized as a concentrated protein supplement for livestock rations?

A. Barley straw
B. Nougcake (Guizotia abyssinica meal)
C. Corn cobs
D. Fresh elephant grass

____ **24.** In poultry production, which viral disease causes acute respiratory distress, nervous tremors, paralysis, high mortality, and requires mandatory vaccination?

A. Newcastle Disease
B. Fowl Coccidiosis
C. Infectious Coryza
D. Scaly Leg Mites

____ **25.** Which agroforestry system combines woody perennial trees with agricultural field crops and pasture animals on the same management unit?

A. Silvopasture / Agrosilvopastoral system
B. Monoculture forestry
C. Hydroponics
D. Open-cast mining

____ **26.** What is the primary objective of post-harvest hermetic storage bags (such as Purdue Improved Crop Storage - PICS bags)?

A. To accelerate seed germination in storage
B. To suffocate storage insect pests by depleting oxygen and trapping CO_2 without chemical fumigants
C. To bleach grains to improve appearance
D. To raise storage humidity to encourage fermentation

____ **27.** In farm accounting, what document records all assets, liabilities, and owners' equity of an agricultural enterprise at a specific point in time?

A. Balance Sheet
B. Income (Profit & Loss) Statement
C. Cash Flow Statement
D. Physical Inventory Log

____ **28.** A parasitic weed that attaches to the roots of sorghum and maize, causing severe chlorosis and yield loss in sub-Saharan Africa, is:

A. Striga hermonthica (Witchweed)
B. Parthenium hysterophorus
C. Lantana camara
D. Eichhornia crassipes

____ **29.** What is the primary purpose of subsoiling or deep ripping in agricultural soil management?

A. To bury crop seeds under 50 cm of soil
B. To fracture dense compacted plow pans and enhance water percolation and root penetration
C. To kill earthworms and microorganisms
D. To bring bedrock stones to the soil surface

____ **30.** Which honeybee caste develops from unfertilized eggs through arrhenotokous parthenogenesis?

A. Drone (male bee)
B. Worker bee (sterile female)
C. Queen bee (fertile female)
D. Royal nurse bee